

Chapter 2.5

Class 5 – Oxidizing substances and organic peroxides

2.5.0 Introductory note

Because of the differing properties exhibited by dangerous goods within classes 5.1 and 5.2, it is impracticable to establish a single criterion for classification in either class. Tests and criteria for assignment to the two classes are addressed in this chapter.

2.5.1 Definitions and general provisions

In this Code, class 5 is divided into two classes as follows:

Class 5.1 – Oxidizing substances

Substances which, while in themselves not necessarily combustible, may, generally by yielding oxygen, cause, or contribute to, the combustion of other material. Such substances may be contained in an article;

Class 5.2 – Organic peroxides

Organic substances which contain the bivalent –O–O– structure and may be considered derivatives of hydrogen peroxide, where one or both of the hydrogen atoms have been replaced by organic radicals. Organic peroxides are thermally unstable substances which may undergo exothermic self-accelerating decomposition. In addition, they may have one or more of the following properties:

- be liable to explosive decomposition;
- burn rapidly;
- be sensitive to impact or friction;
- react dangerously with other substances;
- cause damage to the eyes.

2.5.2 Class 5.1 – Oxidizing substances

Note 1: For the classification of oxidizing substances to class 5.1, in the event of divergence between test results and known experience, judgement based on known experience shall take precedence over test results.

Note 2: By exception, solid ammonium nitrate based fertilizers shall be classified in accordance with the procedure as set out in the *Manual of Tests and Criteria*, part III, section 39.

2.5.2.1 Properties

- 2.5.2.1.1** Substances of class 5.1 in certain circumstances directly or indirectly evolve oxygen. For this reason, oxidizing substances increase the risk and intensity of fire in combustible material with which they come into contact.
- 2.5.2.1.2** Mixtures of oxidizing substances with combustible material and even with material such as sugar, flour, edible oils, mineral oils, etc., are dangerous. These mixtures are readily ignited, in some cases by friction or impact. They may burn violently and may lead to explosion.
- 2.5.2.1.3** There will be a violent reaction between most oxidizing substances and liquid acids, evolving toxic gases. Toxic gases may also be evolved when certain oxidizing substances are involved in a fire.
- 2.5.2.1.4** The above-mentioned properties are, in general, common to all substances in this class. Additionally, some substances possess specific properties, which shall be taken into account in transport. These properties are shown in the Dangerous Goods List in chapter 3.2.

2.5.2.2 Oxidizing solids**2.5.2.2.1 Classification of solid substances of class 5.1**

2.5.2.2.1.1 Tests are performed to measure the potential for the solid substance to increase the burning rate or burning intensity of a combustible substance when the two are thoroughly mixed. The procedure is given in the *Manual of Tests and Criteria*, part III, subsection 34.4.1 (test O.1) or alternatively, in subsection 34.4.3 (test O.3). Tests are conducted on the substance to be evaluated mixed with dry fibrous cellulose in mixing ratios of 1:1 and 4:1, by mass, of sample to cellulose. The burning characteristics of the mixtures are compared:

- .1 in the test O.1, with the standard 3:7 mixture, by mass, of potassium bromate to cellulose. If the burning time is equal to or less than this standard mixture, the burning times shall be compared with those from the packing group I or II reference standards, 3:2 and 2:3 ratios, by mass, of potassium bromate to cellulose, respectively; or
- .2 in the test O.3, with the standard 1:2 mixture, by mass, of calcium peroxide to cellulose. If the burning rate is equal to or greater than this standard mixture, the burning rates shall be compared with those from the packing group I or II reference standards, 3:1 and 1:1 ratios, by mass, of calcium peroxide to cellulose, respectively.

2.5.2.2.1.2 The classification test results are assessed on the basis of:

- .1 the comparison of the mean burning time (for the test O.1) or burning rate (for the test O.3) with those of the reference mixtures; and
- .2 whether the mixture of substance and cellulose ignites and burns.

2.5.2.2.1.3 A solid substance is classified in class 5.1 if the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits:

- .1 in the test O.1, a mean burning time equal to or less than the mean burning time of a 3:7 mixture (by mass) of potassium bromate and cellulose; or
- .2 in the test O.3, a mean burning rate equal to or greater than the mean burning rate of a 1:2 mixture (by mass) of calcium peroxide and cellulose.

2.5.2.2.2 Assignment of packing groups

Solid oxidizing substances are assigned to a packing group according to one of the test procedures in the *Manual of Tests and Criteria*, part III, subsection 34.4.1 (test O.1) or subsection 34.4.3 (test O.3), in accordance with the following criteria:

- .1 Test O.1:
 - .1 Packing group I: any substance which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits a mean burning time less than the mean burning time of a 3:2 mixture, by mass, of potassium bromate and cellulose;
 - .2 Packing group II: any substance which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits a mean burning time equal to or less than the mean burning time of a 2:3 mixture (by mass) of potassium bromate and cellulose, and the criteria for packing group I are not met;
 - .3 Packing group III: any substance which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits a mean burning time equal to or less than the mean burning time of a 3:7 mixture (by mass) of potassium bromate and cellulose, and the criteria for packing groups I and II are not met;
 - .4 Not class 5.1: any substance which, in both the 4:1 and 1:1 sample-to-cellulose ratio (by mass) tested, does not ignite and burn, or exhibits mean burning times greater than that of a 3:7 mixture (by mass) of potassium bromate and cellulose.
- .2 Test O.3:
 - .1 Packing group I: any substance which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits a mean burning rate greater than the mean burning rate of a 3:1 mixture (by mass) of calcium peroxide and cellulose;
 - .2 Packing group II: any substance which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits a mean burning rate equal to or greater than the mean burning rate of a 1:1 mixture (by mass) of calcium peroxide and cellulose, and the criteria for packing group I are not met;
 - .3 Packing group III: any substance which, in the 4:1 or 1:1 sample-to-cellulose ratio (by mass) tested, exhibits a mean burning rate equal to or greater than the mean burning rate of a 1:2 mixture (by mass) of calcium peroxide and cellulose, and the criteria for packing groups I and II are not met;
 - .4 Not class 5.1: any substance which, in both the 4:1 and 1:1 sample-to-cellulose ratio (by mass) tested, does not ignite and burn, or exhibits a mean burning rate less than the mean burning rate of a 1:2 mixture (by mass) of calcium peroxide and cellulose.

Part 2 – Classification**2.5.2.3 Oxidizing liquids****2.5.2.3.1 Classification of liquid substances of class 5.1**

2.5.2.3.1.1 A test is performed to determine the potential for a liquid substance to increase the burning rate or burning intensity of a combustible substance or for spontaneous ignition to occur when the two are thoroughly mixed. The procedure is given in the *Manual of Tests and Criteria*, part III, 34.4.2 (test O.2). It measures the pressure rise time during combustion. Whether a liquid is an oxidizing substance of class 5.1 and, if so, whether packing group I, II or III shall be assigned, is decided on the basis of the test result (see also Precedence of hazard characteristics in 2.0.3).

2.5.2.3.1.2 The classification test results are assessed on the basis of:

- .1 whether the mixture of substance and cellulose spontaneously ignites;
- .2 the comparison of the mean time taken for the pressure to rise from 690 kPa to 2070 kPa gauge with those of the reference substances.

2.5.2.3.1.3 A liquid substance is classified in class 5.1 if the 1:1 mixture, by mass, of substance and cellulose tested exhibits a mean pressure rise time less than or equal to the mean pressure rise time of a 1:1 mixture, by mass, of 65% aqueous nitric acid and cellulose.

2.5.2.3.2 Assignment of packing groups

2.5.2.3.2.1 Liquid oxidizing substances are assigned to a packing group according to the test procedure in the *Manual of Tests and Criteria*, part III, 34.4.2, in accordance with the following criteria:

- .1 Packing group I: any substance which, in the 1:1 mixture (by mass) of substance and cellulose tested, spontaneously ignites; or the mean pressure rise time of a 1:1 mixture (by mass) of substance and cellulose is less than that of a 1:1 mixture (by mass) of 50% perchloric acid and cellulose;
- .2 Packing group II: any substance which, in the 1:1 mixture (by mass) of substance and cellulose tested, exhibits a mean pressure rise time less than or equal to the mean pressure rise time of a 1:1 mixture (by mass) of 40% aqueous sodium chlorate solution and cellulose; and the criteria for packing group I are not met;
- .3 Packing group III: any substance which, in the 1:1 mixture (by mass) of substance and cellulose tested, exhibits a mean pressure rise time less than or equal to the mean pressure rise time of a 1:1 mixture (by mass) of 65% aqueous nitric acid and cellulose; and the criteria for packing groups I and II are not met;
- .4 Not classified as class 5.1: any substance which, in the 1:1 mixture (by mass) of substance and cellulose tested, exhibits a pressure rise of less than 2070 kPa gauge; or exhibits a mean pressure rise time greater than the mean pressure rise time of a 1:1 mixture (by mass) of 65% aqueous nitric acid and cellulose.

2.5.3 Class 5.2 – Organic peroxides**2.5.3.1 Properties**

2.5.3.1.1 Organic peroxides are liable to exothermic decomposition at normal or elevated temperatures. The decomposition can be initiated by heat, contact with impurities (such as acids, heavy-metal compounds, amines), friction or impact. The rate of decomposition increases with temperature and varies with the organic peroxide formulation. Decomposition may result in the evolution of harmful, or flammable, gases or vapours. For certain organic peroxides the temperature shall be controlled during transport. Some organic peroxides may decompose explosively, particularly if confined. This characteristic may be modified by the addition of diluents or by the use of appropriate packagings. Many organic peroxides burn vigorously.

2.5.3.1.2 Contact of organic peroxides with the eyes is to be avoided. Some organic peroxides will cause serious injury to the cornea, even after brief contact, or will be corrosive to the skin.

2.5.3.2 Classification of organic peroxides

2.5.3.2.1 Any organic peroxide shall be considered for classification in class 5.2, unless the organic peroxide formulation contains:

- .1 not more than 1.0% available oxygen from the organic peroxides when containing not more than 1.0% hydrogen peroxide; or
- .2 not more than 0.5% available oxygen from the organic peroxides when containing more than 1.0% but not more than 7.0% hydrogen peroxide.

Note: The available oxygen content (%) of an organic peroxide formulation is given by the formula:

$$16 \times \Sigma(n_i \times c_i / m_i)$$

where:

- n_i = number of peroxygen groups per molecule of organic peroxide i ;
- c_i = concentration (mass %) of organic peroxide i ;
- m_i = molecular mass of organic peroxide i .

- 2.5.3.2.2** Organic peroxides are classified into seven types according to the degree of danger they present. The types of organic peroxide range from type A, which may not be accepted for transport in the packaging in which it is tested, to type G, which is not subject to the provisions for organic peroxides of class 5.2. The classification of types B to F is directly related to the maximum quantity allowed in one packaging.
- 2.5.3.2.3** Organic peroxides permitted for transport in packagings are listed in 2.5.3.2.4, those permitted for transport in IBCs are listed in packing instruction IBC520 and those permitted for transport in portable tanks are listed in portable tank instruction T23. For each permitted substance listed, the generic entry of the Dangerous Goods List (UN 3101 to UN 3120) is assigned, appropriate subsidiary hazards and remarks providing relevant transport information are given. The generic entries specify:
- .1 organic peroxide type (B to F);
 - .2 physical state (liquid or solid); and
 - .3 temperature control, when required (see 2.5.3.4).
- 2.5.3.2.3.1** Mixtures of the listed formulations may be classified as the same type of organic peroxide as that of the most dangerous component and be transported under the conditions of transport given for this type. However, as two stable components can form a thermally less stable mixture, the self-accelerating decomposition temperature (SADT) of the mixture shall be determined and, if necessary, temperature control applied as required by 2.5.3.4.

Part 2 – Classification

2.5.3.2.4 List of currently assigned organic peroxides in packagings

△ **Note:** Packing Method codes “OP1” to “OP8” refer to packing methods in packing instruction P520. Peroxides to be transported shall fulfil the classification and the control and emergency temperatures (derived from the SADT) as listed. For substances permitted in IBCs, see packing instruction IBC520, and for those permitted in tanks, see portable tank instruction T23. The formulations not listed in this subsection but listed in packing instruction IBC520 of 4.1.4.2 and in portable tank instruction T23 of 4.2.5.2.6 may also be transported packed in accordance with packing method OP8 of packing instruction P520 of 4.1.4.1, with the same control and emergency temperatures, if applicable.

Number (generic entry)	ORGANIC PEROXIDE	Concentration (%)	Diluent type A (%)	Diluent type B (%) ⁽¹⁾	Inert solid (%)	Water (%)	Packing method	Control temperature (°C)	Emergency temperature (°C)	Subsidiary hazards and remarks
3101	<i>tert</i> -BUTYL PEROXYACETATE	> 52 – 77	≥ 23				OP5			(3)
	1,1-DI-(<i>tert</i> -BUTYLPEROXY)CYCLOHEXANE	> 80 – 100					OP5			(3)
	1,1-DI-(<i>tert</i> -BUTYLPEROXY)-3,3,5-TRIMETHYLCYCLOHEXANE	> 90 – 100					OP5			(3)
	METHYL ETHYL KETONE PEROXIDE(S)	see remark (8)	≥ 48				OP5			(3) (8) (13)
	2,5-DIMETHYL-2,5-DI-(<i>tert</i> -BUTYLPEROXY)-HEXYNE-3	> 86 – 100					OP5			(3)
	<i>tert</i> -BUTYL MONOPEROXYMALEATE	> 52 – 100					OP5			(3)
3102	3-CHLOROPEROXYBENZOIC ACID	> 57 – 86			≥ 14		OP1			(3)
	DIBENZOYL PEROXIDE	> 52 – 100			≤ 48		OP2			(3)
	DIBENZOYL PEROXIDE	> 77 – 94				≥ 6	OP4			(3)
	DI-4-CHLOROBENZOYL PEROXIDE	≤ 77				≥ 23	OP5			(3)
	DI-2,4-DICHLOROBENZOYL PEROXIDE	≤ 77				≥ 23	OP5			(3)
	2,2-DIHYDROPEROXYPROPANE	≤ 27					OP5			(3)
	2,5-DIMETHYL-2,5-DI-(BENZOYLPEROXY)HEXANE	> 82 – 100			≥ 73		OP5			(3)
	DI-(2-PHENOXYETHYL) PEROXYDICARBONATE	> 85 – 100					OP5			(3)
	DISUCCINIC ACID PEROXIDE	> 72 – 100					OP4			(3) (17)
	<i>tert</i> -AMYL PEROXYBENZOATE	≤ 100					OP5			
3103	<i>tert</i> -AMYLPEROXY ISOPROPYL CARBONATE	≤ 77	≥ 23				OP5			
	<i>n</i> -BUTYL 4,4-DI-(<i>tert</i> -BUTYLPEROXY)VALERATE	> 52 – 100					OP5			
	<i>tert</i> -BUTYL HYDROPEROXIDE	> 79 – 90				≥ 10	OP5			(13)
	<i>tert</i> -BUTYL HYDROPEROXIDE + DI- <i>tert</i> -BUTYL PEROXIDE	< 82 + > 9				≥ 7	OP5			(13)
	<i>tert</i> -BUTYL MONOPEROXYMALEATE	≤ 52	≥ 48				OP6			
	<i>tert</i> -BUTYL PEROXYACETATE	> 32 – 52	≥ 48				OP6			
	<i>tert</i> -BUTYL PEROXYBENZOATE	> 77 – 100					OP5			
	<i>tert</i> -BUTYLPEROXY ISOPROPYLCARBONATE	≤ 77	≥ 23				OP5			
	<i>tert</i> -BUTYLPEROXY-2-METHYLBENZOATE	≤ 100					OP5			

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Number (generic entry)	ORGANIC PEROXIDE	Concentration (%)	Diluent type A (%)	Diluent type B (%) ⁽¹⁾	Inert solid (%)	Water (%)	Packing method	Control temperature (°C)	Emergency temperature (°C)	Subsidiary hazards and remarks
3103 (cont.)	1,1-DI-(<i>tert</i> -AMYLPEROXY)CYCLOHEXANE	≤ 82	≥ 18				OP6			
	2,2-DI-(<i>tert</i> -BUTYLPEROXY)BUTANE	≤ 52	≥ 48				OP6			
	1,6-DI-(<i>tert</i> -BUTYLPEROXYCARBONYLOXY)-HEXANE	≤ 72	≥ 28				OP5			
	1,1-DI-(<i>tert</i> -BUTYLPEROXY)CYCLOHEXANE	> 52 – 80	≥ 20				OP5			
	1,1-DI-(<i>tert</i> -BUTYLPEROXY)CYCLOHEXANE	≤ 72		≥ 28			OP5			(30)
	1,1-DI-(<i>tert</i> -BUTYLPEROXY)-3,3,5-TRIMETHYL-CYCLOHEXANE	> 57 – 90	≥ 10				OP5			
	1,1-DI-(<i>tert</i> -BUTYLPEROXY)-3,3,5-TRIMETHYL-CYCLOHEXANE	≤ 77		≥ 23			OP5			
	1,1-DI-(<i>tert</i> -BUTYLPEROXY)-3,3,5-TRIMETHYL-CYCLOHEXANE	≤ 90		≥ 10			OP5			(30)
	2,5-DIMETHYL-2,5-DI-(<i>tert</i> -BUTYLPEROXY)HEXANE	> 90 – 100					OP5			
	2,5-DIMETHYL-2,5-DI-(<i>tert</i> -BUTYLPEROXY)-HEXYNE-3	> 52 – 86	≥ 14				OP5			(26)
	ETHYL 3,3-DI-(<i>tert</i> -BUTYLPEROXY)BUTYRATE	> 77 – 100					OP5			
	ORGANIC PEROXIDE, LIQUID, SAMPLE						OP2			(11)
	CYCLOHEXANONE PEROXIDE(S)	≤ 91				≥ 9	OP6			(13)
3104	DI-2,4-DICHLOROBENZOYL PEROXIDE	≤ 52 as a paste with silicon oil					OP5			
	DIBENZOYL PEROXIDE	≤ 77				≥ 23	OP6			
	2,5-DIMETHYL-2,5-DI(BENZOYLPEROXY)HEXANE	≤ 82				≥ 18	OP5			
	2,5-DIMETHYL-2,5-DIHYDROPEROXYHEXANE	≤ 82				≥ 18	OP6			
	ORGANIC PEROXIDE, SOLID, SAMPLE						OP2			(11)
	ACETYL ACETONE PEROXIDE	≤ 42	≥ 48			≥ 8	OP7			(2)
3105	<i>tert</i> -AMYL PEROXYACETATE	≤ 62	≥ 38				OP7			
	<i>tert</i> -AMYL PEROXY-2-ETHYLHEXYL CARBONATE	≤ 100					OP7			
	<i>tert</i> -AMYL PEROXY-3,5,5-TRIMETHYLHEXANOATE	≤ 100					OP7			
	<i>tert</i> -BUTYL HYDROPEROXIDE	≤ 80	≥ 20				OP7			(4) (13)
	<i>tert</i> -BUTYL PEROXYBENZOATE	> 52 – 77	≥ 23				OP7			
	<i>tert</i> -BUTYL PEROXYBUTYL FUMARATE	≤ 52	≥ 48				OP7			
	<i>tert</i> -BUTYL PEROXYCROTONATE	≤ 77	≥ 23				OP7			
	<i>tert</i> -BUTYL PEROXY-2-ETHYLHEXYLCARBONATE	≤ 100					OP7			
	<i>tert</i> -BUTYLPEROXY ISOPROPYLCARBONATE	≤ 62		≥ 38			OP7			

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Part 2 – Classification

Number (generic entry)	ORGANIC PEROXIDE	Concentration (%)	Diluent type A (%)	Diluent type B (%) ⁽¹⁾	Inert solid (%)	Water (%)	Packing method	Control temperature (°C)	Emergency temperature (°C)	Subsidiary hazards and remarks
3105 (cont.)	1-(2- <i>tert</i> -BUTYLPEROXY ISOPROPYL)-3-ISOPROPENYLBENZENE	≤ 77	≥ 23				OP7			
	<i>tert</i> -BUTYL PEROXY-3,5,5-TRIMETHYLHEXANOATE	> 37 – 100					OP7			
	CYCLOHEXANONE PEROXIDE(S)	≤ 72	≥ 28				OP7			(5)
	2,2-DI-(<i>tert</i> -AMYLPEROXY)BUTANE	≤ 57	≥ 43				OP7			
	DI- <i>tert</i> -BUTYL PEROXYAZELATE	≤ 52	≥ 48				OP7			
	1,1-DI-(<i>tert</i> -BUTYLPEROXY)CYCLOHEXANE	> 42 – 52	≥ 48				OP7			
	1,1-DI-(<i>tert</i> -BUTYLPEROXY)CYCLOHEXANE + <i>tert</i> -BUTYL PEROXY-2-ETHYLHEXANOATE	≤ 43 + ≤ 16	≥ 41				OP7			
	DI-(<i>tert</i> -BUTYLPEROXY)PHTHALATE	> 42 – 52	≥ 48				OP7			
	2,2-DI-(<i>tert</i> -BUTYLPEROXY)PROPANE	≤ 52	≥ 48				OP7			
	2,5-DIMETHYL-2,5-DI-(<i>tert</i> -BUTYLPEROXY)HEXANE	> 52 – 90	≥ 10				OP7			
	2,5-DIMETHYL-2,5-DI-(3,5,5-TRIMETHYL- HEXANOYLPEROXY)HEXANE	≤ 77	≥ 23				OP7			
	ETHYL 3,3-DI-(<i>tert</i> -AMYLPEROXY)BUTYRATE	≤ 67	≥ 33				OP7			
	ETHYL 3,3-DI-(<i>tert</i> -BUTYLPEROXY)BUTYRATE	≤ 77	≥ 23				OP7			(13)
	<i>p</i> -MENTHYL HYDROPEROXIDE	> 72 – 100					OP7			(9)
	METHYL ETHYL KETONE PEROXIDE(S)	see remark (9)	≥ 55				OP7			(33) (34)
	METHYL ETHYL KETONE PEROXIDE(S)	see remark (33)	≥ 41			≥ 9	OP8			(22)
3106	METHYL ISOBUTYL KETONE PEROXIDE(S)	≤ 62	≥ 19				OP7			(13) (14) (19)
	PEROXYACETIC ACID, TYPE D, stabilized	≤ 43					OP7			(13)
	PINANYL HYDROPEROXIDE	> 56 – 100					OP7			(28)
	1,1,3,3-TETRAMETHYLBUTYL HYDROPEROXIDE	≤ 100					OP7			
	3,6,9-TRIETHYL-3,6,9-TRIMETHYL-1,4,7- TRIPEROXONANE	≤ 42	≥ 58				OP7			
	ACETYL ACETONE PEROXIDE	≤ 32 as a paste					OP7			(20)
	<i>tert</i> -BUTYL PEROXYBENZOATE	≤ 52			≥ 48		OP7			
	<i>tert</i> -BUTYL PEROXY-2-ETHYLHEXANOATE + 2,2-DI-(<i>tert</i> -BUTYLPEROXY)BUTANE	≤ 12 + ≤ 14	≥ 14		≥ 60		OP7			
	<i>tert</i> -BUTYLPEROXY STEARYLCARBONATE	≤ 100					OP7			
	<i>tert</i> -BUTYL PEROXY-3,5,5-TRIMETHYLHEXANOATE	≤ 42			≥ 58		OP7			
	3-CHLOROPEROXYBENZOIC ACID	≤ 57			≥ 3	≥ 40	OP7			
	3-CHLOROPEROXYBENZOIC ACID	≤ 77			≥ 6	≥ 17	OP7			(5) (20)
	CYCLOHEXANONE PEROXIDE(S)	≤ 72 as a paste					OP7			
	DIBENZOYL PEROXIDE	≤ 62			≥ 28	≥ 10	OP7			

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Number (generic entry)	ORGANIC PEROXIDE	Concentration (%)	Diluent type A (%)	Diluent type B (%) ⁽¹⁾	Inert solid (%)	Water (%)	Packing method	Control temperature (°C)	Emergency temperature (°C)	Subsidiary hazards and remarks
3106 (cont.)	DIBENZOYL PEROXIDE	> 52 – 62 as a paste					OP7			(20)
	DIBENZOYL PEROXIDE	> 35 – 52			≥ 48		OP7			
	1,1-DI-(<i>tert</i> -BUTYLPEROXY)CYCLOHEXANE	≤ 42	≥ 13		≥ 45		OP7			
	DI-(<i>tert</i> -BUTYLPEROXYISOPROPYL)BENZENE(S)	> 42 – 100			≤ 57		OP7			(20)
	DI-(<i>tert</i> -BUTYLPEROXY)PHTHALATE	≤ 52 as a paste					OP7			
	2,2-DI-(<i>tert</i> -BUTYLPEROXY)PROPANE	≤ 42	≥ 13		≥ 45		OP7			(20)
	DI-4-CHLOROBENZOYL PEROXIDE	≤ 52 as a paste					OP7			
	2,2-DI-(4,4-DI-(<i>tert</i> -BUTYLPEROXY)CYCLOHEXYL)-PROPANE	≤ 42			≥ 58		OP7			
	DI-(1-HYDROXYCYCLOHEXYL)PEROXIDE	≤ 100					OP7			
	DIISOPROPYL BENZENE DIHYDROPEROXIDE	≤ 82	≥ 5			≥ 5	OP7			(24)
	DILAULOYL PEROXIDE	≤ 100					OP7			
	DI-(4-METHYLBENZOYL) PEROXIDE	≤ 52 as paste with silicon oil					OP7			
	2,5-DIMETHYL-2,5-DI-(BENZOYLPEROXY)HEXANE	≤ 82			≥ 18		OP7			
	2,5-DIMETHYL-2,5-DI-(<i>tert</i> -BUTYLPEROXY)-HEXYNE-3	≤ 52			≥ 48		OP7			
	DI-(2-PHENOXYETHYL)PEROXYDICARBONATE	≤ 85				≥ 15	OP7			
	ETHYL 3,3-DI-(<i>tert</i> -BUTYLPEROXY)BUTYRATE	≤ 52			≥ 48		OP7			
	([3R-(3R,5aS,6S,8aS,9R,10R,12S,12aR*)]-DECAHYDRO-10-METHOXY-3,6,9-TRIMETHYL-3,12-EPOXY-12H-PYRANO[4,3- <i>j</i>]-1,2-BENZODIOXEPIN)	≤ 100					OP7			
3107	ACETYL ACETONE PEROXIDE	≤ 35	≥ 57			≥ 8	OP8			(32)
	<i>tert</i> -AMYL HYDROPEROXIDE	≤ 88	≥ 6			≥ 6	OP8			
	<i>tert</i> -BUTYL HYDROPEROXIDE	≤ 79				> 14	OP8			(13) (23)
	CUMYL HYDROPEROXIDE	> 90 – 98	≤ 10				OP8			(13)
	DI- <i>tert</i> -AMYL PEROXIDE	≤ 100					OP8			
	DIBENZOYL PEROXIDE	> 36 – 42	≥ 18			≤ 40	OP8			
	DI- <i>tert</i> -BUTYL PEROXIDE	> 52 – 100					OP8			
	1,1-DI-(<i>tert</i> -BUTYLPEROXY)CYCLOHEXANE	≤ 27	≥ 25				OP8			(21)
	DI-(<i>tert</i> -BUTYLPEROXY)PHTHALATE	≤ 42	≥ 58				OP8			
	1,1-DI-(<i>tert</i> -BUTYLPEROXY)-3,3,5-TRIMETHYLCYCLOHEXANE	≤ 57	≥ 43				OP8			
	1,1-DI-(<i>tert</i> -BUTYLPEROXY)-3,3,5-TRIMETHYLCYCLOHEXANE	≤ 32	≥ 26	≥ 42			OP8			

Part 2 – Classification

Number (generic entry)	ORGANIC PEROXIDE	Concentration (%)	Diluent type A (%)	Diluent type B (%) ⁽¹⁾	Inert solid (%)	Water (%)	Packing method	Control temperature (°C)	Emergency temperature (°C)	Subsidiary hazards and remarks
3107 (cont.)	2,2-DI-(4,4-DI-(<i>tert</i> -BUTYLPEROXY) CYCLOHEXYL)-PROPANE	≤ 22		≥ 78			OP8			
	METHYL ETHYL KETONE PEROXIDE(S)	see remark (10)	≥ 60				OP8			(10)
	3,3,5,7,7-PENTAMETHYL-1,2,4-TRIOXEPANE	≤ 100					OP8			
	PEROXYACETIC ACID, TYPE E, stabilized	≤ 43					OP8			(13) (15) (19)
	POLYETHER POLY- <i>tert</i> -BUTYLPEROXY- CARBONATE	≤ 52		≥ 48			OP8			
3108	<i>tert</i> -BUTYL CUMYL PEROXIDE	≤ 52			≥ 48		OP8			
	<i>n</i> -BUTYL 4,4-DI-(<i>tert</i> -BUTYLPEROXY)VALERATE	≤ 52			≥ 48		OP8			
	<i>tert</i> -BUTYL MONOPEROXYMALEATE	≤ 52			≥ 48		OP8			
	<i>tert</i> -BUTYL MONOPEROXYMALEATE	≤ 52 as a paste					OP8			
	1-(2- <i>tert</i> -BUTYLPEROXYISOPROPYL)-3- ISOPROPENYLBENZENE	≤ 42			≥ 58		OP8			
	DIBENZOYL PEROXIDE	≤ 56.5 as a paste				≥ 15	OP8			
	DIBENZOYL PEROXIDE	≤ 52 as a paste					OP8			(20)
	2,5-DIMETHYL-2,5-DI-(<i>tert</i> -BUTYLPEROXY)HEXANE	≤ 47 as a paste					OP8			
	2,5-DIMETHYL-2,5-DI-(<i>tert</i> -BUTYLPEROXY)HEXANE	≤ 77			≥ 23		OP8			
	<i>tert</i> -BUTYL CUMYL PEROXIDE	> 42 – 100					OP8			
3109	<i>tert</i> -BUTYL HYDROPEROXIDE	≤ 72				≥ 28	OP8			(13)
	<i>tert</i> -BUTYL PEROXYACETATE	≤ 32		≥ 68			OP8			
	<i>tert</i> -BUTYL PEROXY-3,5,5-TRIMETHYL-HEXANOATE	≤ 37		≥ 63			OP8			
	CUMYL HYDROPEROXIDE	≤ 90	≥ 10				OP8			(13) (18)
	DIBENZOYL PEROXIDE	≤ 42	≥ 38			≥ 13	OP8			
	DIBENZOYL PEROXIDE	≤ 42 as a stable dispersion in water					OP8			
	DI- <i>tert</i> -BUTYL PEROXIDE	≤ 52		≥ 48			OP8			(25)
	1,1-DI-(<i>tert</i> -BUTYLPEROXY)CYCLOHEXANE	≤ 42	≥ 58				OP8			
	1,1-DI-(<i>tert</i> -BUTYLPEROXY)CYCLOHEXANE	≤ 13	≥ 13	≥ 74			OP8			
	DILAULOYL PEROXIDE	≤ 42 as a stable dispersion in water					OP8			
	2,5-DIMETHYL-2,5-DI-(<i>tert</i> -BUTYLPEROXY)HEXANE	≤ 52	≥ 48				OP8			
	ISOPROPYLCUMYL HYDROPEROXIDE	≤ 72	≥ 28				OP8			(13)
	<i>p</i> -MENTHYL HYDROPEROXIDE	≤ 72	≥ 28				OP8			(27)
	METHYL ISOPROPYL KETONE PEROXIDE(S)	See remark (31)	≥ 70				OP8			(31)
	PEROXYACETIC ACID, TYPE F, stabilized	≤ 43					OP8			(13) (16) (19)

Chapter 2.5 – Class 5 – Oxidizing substances and organic peroxides

Number (generic entry)	ORGANIC PEROXIDE	Concentration (%)	Diluent type A (%)	Diluent type B (%) ⁽¹⁾	Inert solid (%)	Water (%)	Packing method	Control temperature (°C)	Emergency temperature (°C)	Subsidiary hazards and remarks
3109 (cont.)	1-PHENYLETHYL HYDROPEROXIDE	≤ 38		≥ 62			OP8			
	PINANYL HYDROPEROXIDE	≤ 56	≥ 44				OP8			
3110	DICUMYL PEROXIDE	> 52 – 100					OP8			(12)
	1,1-DI-(<i>tert</i> -BUTYLPEROXY)-3,3,5-TRIMETHYL- CYCLOHEXANE	≤ 57			≥ 43		OP8			
	3,6,9-TRIETHYL-3,6,9-TRIMETHYL-1,4,7- TRIPEROXONANE	≤ 17	≥ 18		≥ 65		OP8			
3111	<i>tert</i> -BUTYL PEROXYISOBUTYRATE	> 52 – 77		≥ 23			OP5	+15	+20	(3)
	DIISOBUTYRYL PEROXIDE	> 32 – 52		≥ 48			OP5	-20	-10	(3)
	ISOPROPYL <i>sec</i> -BUTYL PEROXYDICARBONATE + DI- <i>sec</i> -BUTYL PEROXYDICARBONATE + DIISOPROPYL PEROXYDICARBONATE	≤ 52 + ≤ 28 + ≤ 22					OP5	-20	-10	(3)
	ACETYL CYCLOHEXANESULFONYL PEROXIDE	≤ 82				≥ 12	OP4	-10	0	(3)
3112	DICYCLOHEXYL PEROXYDICARBONATE	> 91 – 100					OP3	+10	+15	(3)
	DIISOPROPYL PEROXYDICARBONATE	> 52 – 100					OP2	-15	-5	(3)
	DI-(2-METHYLBENZOYL) PEROXIDE	≤ 87				≥ 13	OP5	+30	+35	(3)
	<i>tert</i> -AMYL PEROXYPIVALATE	≤ 77		≥ 23			OP5	+10	+15	
3113	<i>tert</i> -BUTYL PEROXYDIETHYLACETATE	≤ 100					OP5	+20	+25	
	<i>tert</i> -BUTYL PEROXY-2-ETHYLHEXANOATE	> 52 – 100					OP6	+20	+25	
	<i>tert</i> -BUTYL PEROXYPIVALATE	> 67 – 77	≥ 23				OP5	0	+10	
	DI- <i>sec</i> -BUTYL PEROXYDICARBONATE	> 52 – 100					OP4	-20	-10	
	DI-(2-ETHYLHEXYL)PEROXYDICARBONATE	> 77 – 100					OP5	-20	-10	
	2,5-DIMETHYL-2,5-DI-(2-ETHYLHEXANOYLPEROXY)- HEXANE	≤ 100					OP5	+20	+25	
	DI- <i>n</i> -PROPYL PEROXYDICARBONATE	≤ 100					OP3	-25	-15	
	DI- <i>n</i> -PROPYL PEROXYDICARBONATE	≤ 77		≥ 23			OP5	-20	-10	(11)
3114	ORGANIC PEROXIDE, LIQUID, SAMPLE, TEMPERATURE CONTROLLED						OP2			
	DI-(4- <i>tert</i> -BUTYLCYCLOHEXYL)- PEROXYDICARBONATE	≤ 100					OP6	+30	+35	
	DICYCLOHEXYL PEROXYDICARBONATE	≤ 91			≥ 9		OP5	+10	+15	
	DIDECANOYL PEROXIDE	≤ 100					OP6	+30	+35	
	DI- <i>n</i> -OCTANOYL PEROXIDE	≤ 100					OP5	+10	+15	
	ORGANIC PEROXIDE, SOLID, SAMPLE, TEMPERATURE CONTROLLED						OP2			(11)

Part 2 – Classification

Number (generic entry)	ORGANIC PEROXIDE	Concentration (%)	Diluent type A (%)	Diluent type B (%) ⁽¹⁾	Inert solid (%)	Water (%)	Packing method	Control temperature (°C)	Emergency temperature (°C)	Subsidiary hazards and remarks
3115	ACETYL CYCLOHEXANESULPHONYL PEROXIDE	≤ 32		≥ 68			OP7	-10	0	
	tert-AMYL PEROXY-2-ETHYLHEXANOATE	≤ 100					OP7	+20	+25	
	tert-AMYL PEROXYNEODECANOATE	≤ 77		≥ 23			OP7	0	+10	
	tert-BUTYL PEROXY-2-ETHYLHEXANOATE + 2,2-DI-(tert-BUTYLPEROXY)BUTANE	≤ 31 + ≤ 36		≥ 33			OP7	+35	+40	
	tert-BUTYL PEROXYISOBUTYRATE	≤ 52		≥ 48			OP7	+15	+20	
	tert-BUTYL PEROXYNEODECANOATE	> 77 – 100					OP7	-5	+5	
	tert-BUTYL PEROXYNEODECANOATE	≤ 77		≥ 23			OP7	0	+10	
	tert-BUTYL PEROXYNEOHEPTANOATE	≤ 77	≥ 23				OP7	0	+10	
	tert-BUTYL PEROXYPIVALATE	> 27 – 67		≥ 33			OP7	0	+10	
	CUMYL PEROXYNEODECANOATE	≤ 77		≥ 23			OP7	-10	0	
	CUMYL PEROXYNEODECANOATE	≤ 87	≥ 13				OP7	-10	0	
	CUMYL PEROXYNEOHEPTANOATE	≤ 77	≥ 23				OP7	-10	0	
	CUMYL PEROXYPIVALATE	≤ 77		≥ 23			OP7	-5	+5	
	DIACETONE ALCOHOL PEROXIDES	≤ 57		≥ 26		≥ 8	OP7	+40	+45	(6)
	DIACETYL PEROXIDE	≤ 27		≥ 73			OP7	+20	+25	(7) (13)
	DI-n-BUTYL PEROXYDICARBONATE	> 27 – 52		≥ 48			OP7	-15	-5	
	DI-sec-BUTYL PEROXYDICARBONATE	≤ 52		≥ 48			OP7	-15	-5	
	DI-(2-ETHOXYETHYL)PEROXYDICARBONATE	≤ 52		≥ 48			OP7	-10	0	
	DI-(2-ETHYLHEXYL)PEROXYDICARBONATE	≤ 77		≥ 23			OP7	-15	-5	
	DIISOBUTYRYL PEROXIDE	≤ 32		≥ 68			OP7	-20	-10	
	DIISOPROPYL PEROXYDICARBONATE	≤ 52		≥ 48			OP7	-20	-10	
	DIISOPROPYL PEROXYDICARBONATE	≤ 32	≥ 68				OP7	-15	-5	
	DI-(3-METHOXYBUTYL) PEROXYDICARBONATE	≤ 52		≥ 48			OP7	-5	+5	
	DI-(3-METHYLBENZOYL) PEROXIDE + BENZOYL (3-METHYLBENZOYL) PEROXIDE + DIBENZOYL PEROXIDE	≤ 20 + ≤ 18 + ≤ 4		≥ 58			OP7	+35	+40	
	DI-(2-NEODECANOYL)PEROXYISOPROPYL)- BENZENE	≤ 52	≥ 48				OP7	-10	0	
	DI-(3,5,5-TRIMETHYLHEXANOYL) PEROXIDE	> 52 – 82	≥ 18				OP7	0	+10	
	1-(2-ETHYLHEXANOYL)PEROXY)-1,3- DIMETHYLBUTYL PEROXYPIVALATE	≤ 52	≥ 45	≥ 10			OP7	-20	-10	
	tert-HEXYL PEROXYNEODECANOATE	≤ 71	≥ 29				OP7	0	+10	
	tert-HEXYL PEROXYPIVALATE	≤ 72		≥ 28			OP7	+10	+15	

Chapter 2.5 – Class 5 – Oxidizing substances and organic peroxides

Number (generic entry)	ORGANIC PEROXIDE	Concentration (%)	Diluent type A (%)	Diluent type B (%) ⁽¹⁾	Inert solid (%)	Water (%)	Packing method	Control temperature (°C)	Emergency temperature (°C)	Subsidiary hazards and remarks
3115 (cont.)	3-HYDROXY-1,1-DIMETHYLBUTYL PEROXYNEODECANOATE	≤ 77	≥ 23				OP7	-5	+5	
	ISOPROPYL <i>sec</i> -BUTYL PEROXYDICARBONATE + DI- <i>sec</i> -BUTYL PEROXYDICARBONATE + DI-ISOPROPYL PEROXYDICARBONATE	≤ 32 + ≤ 15 - 18 + ≤ 12 - 15	≥ 38				OP7	-20	-10	
	METHYLCYCLOHEXANONE PEROXIDE(S)	≤ 67		≥ 33			OP7	+35	+40	
	1,1,3,3-TETRAMETHYLBUTYL PEROXY-2-ETHYLHEXANOATE	≤ 100					OP7	+15	+20	
	1,1,3,3-TETRAMETHYLBUTYL PEROXY-NEODECANOATE	≤ 72		≥ 28			OP7	-5	+5	
3116	1,1,3,3-TETRAMETHYLBUTYL PEROXYPIVALATE	≤ 77	≥ 23				OP7	0	+10	
	DIMYRISTYL PEROXYDICARBONATE	≤ 100					OP7	+20	+25	
	DI- <i>n</i> -NONANOYL PEROXIDE	≤ 100					OP7	0	+10	
	DISUCCINIC ACID PEROXIDE	≤ 72				≥ 28	OP7	+10	+15	
3117	<i>tert</i> -BUTYL PEROXY-2-ETHYLHEXANOATE	> 32 - 52		≥ 48			OP8	+30	+35	
	DI- <i>n</i> -BUTYL PEROXYDICARBONATE	≤ 27		≥ 73			OP8	-10	0	
	<i>tert</i> -BUTYL PEROXYNEOHEPTANOATE	≤ 42 as a stable dispersion in water					OP8	0	+10	
	1,1-DIMETHYL-3-HYDROXYBUTYL PEROXY-NEOHEPTANOATE	≤ 52	≥ 48				OP8	0	+10	
	DIPROPIONYL PEROXIDE	≤ 27		≥ 73			OP8	+15	+20	
3118	<i>tert</i> -HEXYL PEROXYPIVALATE	≤ 52 as a stable dispersion in water					OP8	+15	+20	
	3-HYDROXY-1,1-DIMETHYLBUTYL PEROXY-NEODECANOATE	≤ 52	≥ 48				OP8	-5	+5	
	<i>tert</i> -BUTYL PEROXY-2-ETHYLHEXANOATE	≤ 52			≥ 48		OP8	+20	+25	
	<i>tert</i> -BUTYL PEROXYNEODECANOATE	≤ 42 as a stable dispersion in water (frozen)					OP8	0	+10	
	DI-(4- <i>tert</i> -BUTYLCYCLOHEXYL) PEROXYDICARBONATE	≤ 42 as a paste					OP8	+35	+40	
	DI- <i>n</i> -BUTYL PEROXYDICARBONATE	≤ 42 as a stable dispersion in water (frozen)					OP8	-15	-5	
	DI-2,4-DICHLOROBENZOYL PEROXIDE	≤ 52 as a paste					OP8	+20	+25	
	PEROXYLAURIC ACID	≤ 100					OP8	+35	+40	

Part 2 – Classification

Number (generic entry)	ORGANIC PEROXIDE	Concentration (%)	Diluent type A (%)	Diluent type B (%) ⁽¹⁾	Inert solid (%)	Water (%)	Packing method	Control temperature (°C)	Emergency temperature (°C)	Subsidiary hazards and remarks
3119	tert-AMYL PEROXYNEODECANOATE	≤ 47	≥ 53				OP8	0	+ 10	
	tert-BUTYL PEROXY-2-ETHYLHEXANOATE	≤ 32		≥ 68			OP8	+40	+45	
	tert-BUTYL PEROXYNEODECANOATE	≤ 52 as a stable dispersion in water					OP8	0	+10	
	tert-BUTYL PEROXYNEODECANOATE	≤ 32	≥ 68				OP8	0	+10	
	tert-BUTYL PEROXYPIVALATE	≤ 27		≥ 73			OP8	+30	+35	
	CUMYL PEROXYNEODECANOATE	≤ 52 as a stable dispersion in water					OP8	-10	0	
	DI-(4-tert-BUTYLCYCLOHEXYL) PEROXYDICARBONATE	≤ 42 as a stable dispersion in water					OP8	+30	+35	
	DICETYL PEROXYDICARBONATE	≤ 42 as a stable dispersion in water					OP8	+30	+35	
	DICYCLOHEXYL PEROXYDICARBONATE	≤ 42 as a stable dispersion in water					OP8	+15	+20	
	DI-(2-ETHYLHEXYL) PEROXYDICARBONATE	≤ 62 as a stable dispersion in water					OP8	-15	-5	
	DIISOBUTYRYL PEROXIDE	≤ 42 as a stable dispersion in water					OP8	-20	-10	
	DIMYRISTYL PEROXYDICARBONATE	≤ 42 as a stable dispersion in water					OP8	+20	+25	
	DI-(3,5,5-TRIMETHYLHEXANOYL) PEROXIDE	≤ 52 as a stable dispersion in water					OP8	+10	+15	
	DI-(3,5,5-TRIMETHYLHEXANOYL) PEROXIDE	≤ 38	≥ 62				OP8	+20	+25	
3120	DI-(3,5,5-TRIMETHYLHEXANOYL) PEROXIDE	> 38 – 52	≥ 48				OP8	+10	+15	
	3-HYDROXY-1,1-DIMETHYLBUTYL PEROXYNEODECANOATE	≤ 52 as a stable dispersion in water					OP 8	-5	+ 5	
	1,1,3,3-TETRAMETHYLBUTYL PEROXYNEODECANOATE	≤ 52 as a stable dispersion in water					OP8	-5	+5	
	DI-(2-ETHYLHEXYL)PEROXYDICARBONATE	≤ 52 as a stable dispersion in water (frozen)					OP8	-15	-5	
	DICETYL PEROXYDICARBONATE	≤ 100					OP8	+30	+35	
	CYCLOHEXANONE PEROXIDE(S)	≤ 32			≥ 68					(29)
	DIBENZOYL PEROXIDE	≤ 35			≥ 65					(29)
	DI-(tert-BUTYLPEROXYISOPROPYL)BENZENE(S)	≤ 42			≥ 58					(29)
	DI-4-CHLOROBENZOYL PEROXIDE	≤ 32			≥ 68					(29)
	DICUMYL PEROXIDE	≤ 52			≥ 48					(29)
	2,5-DIMETHYL-2,5-DI-(tert-BUTYLPEROXY) HEXANE	≤ 22			≥ 78					(29)
	Exempt									
	Exempt									
	Exempt									
	Exempt									

Remarks

- (1) Diluent type B may always be replaced by diluent type A. The boiling point of diluent type B shall be at least 60°C higher than the SADT of the organic peroxide
- (2) Available oxygen $\leq 4.7\%$
- (3) "EXPLOSIVE" subsidiary hazard label required. (Model No. 1, see 5.2.2.2.2)
- (4) Diluent may be replaced by di-*tert*-butyl peroxide
- (5) Available oxygen $\leq 9\%$
- (6) With $\leq 9\%$ hydrogen peroxide; available oxygen $\leq 10\%$
- (7) Only non-metallic packagings are allowed
- (8) Available oxygen $> 10\%$ and $\leq 10.7\%$, with or without water
- (9) Available oxygen $\leq 10\%$, with or without water
- (10) Available oxygen $\leq 8.2\%$, with or without water
- (11) See 2.5.3.2.5.1
- (12) Up to 2,000 kg per receptacle assigned to ORGANIC PEROXIDE TYPE F on the basis of large-scale trials
- (13) "CORROSIVE" subsidiary hazard label required (Model No. 8, see 5.2.2.2.2)
- (14) Peroxyacetic acid formulations which fulfil the criteria of 2.5.3.3.2.4
- (15) Peroxyacetic acid formulations which fulfil the criteria of 2.5.3.3.2.5
- (16) Peroxyacetic acid formulations which fulfil the criteria of 2.5.3.3.2.6
- (17) Addition of water to this organic peroxide will decrease its thermal stability
- (18) No "CORROSIVE" subsidiary hazard label required for concentrations below 80%
- (19) Mixtures with hydrogen peroxide, water and acid(s)
- (20) With diluent type A, with or without water
- (21) With $\geq 25\%$ diluent type A by mass, and in addition ethylbenzene
- (22) With $\geq 19\%$ diluent type A by mass, and in addition methyl isobutyl ketone
- (23) With $< 6\%$ di-*tert*-butyl peroxide
- (24) With $\leq 8\%$ 1-isopropylhydroperoxy-4-isopropylhydroxybenzene
- (25) Diluent type B with boiling point $> 110^\circ\text{C}$
- (26) With $< 0.5\%$ hydroperoxides content
- (27) For concentrations more than 56%, "CORROSIVE" subsidiary hazard label required (Model No. 8, see 5.2.2.2.2)
- (28) Available active oxygen $\leq 7.6\%$ in diluent type A having a 95% boil-off point in the range 200–260°C
- (29) Not subject to the provisions for peroxide, class 5.2
- (30) Diluent type B with boiling point $> 130^\circ\text{C}$
- (31) Active oxygen $\leq 6.7\%$
- (32) Active oxygen $\leq 4.15\%$
- (33) Available oxygen $\leq 10\%$
- (34) Sum of diluent type A and water $\geq 55\%$, and in addition methyl ethyl ketone

2.5.3.2.5 Classification of organic peroxides not listed in 2.5.3.2.4, packing instruction IBC520 or portable tank instruction T23 and assignment to a generic entry shall be made by the competent authority of the country of origin on the basis of a test report. Principles applying to the classification of such substances are provided in 2.5.3.3. Test methods and criteria and an example of a report are given in the current edition of the *Manual of Tests and Criteria*, part II. The statement of approval shall contain the classification and the relevant transport conditions (see 5.4.4.1.3).

2.5.3.2.5.1 Samples of new organic peroxides or new formulations of currently assigned organic peroxides for which complete test data are not available and which are to be transported for further testing or evaluation may be assigned to one of the appropriate entries for ORGANIC PEROXIDE TYPE C provided the following conditions are met:

- .1 the available data indicate that the sample would be no more dangerous than ORGANIC PEROXIDE TYPE B;
- .2 the sample is packaged in accordance with packing method OP2 and the quantity per cargo transport unit is limited to 10 kg; and
- .3 the available data indicate that the control temperature, if any, is sufficiently low to prevent any dangerous decomposition and sufficiently high to prevent any dangerous phase separation.

2.5.3.3 Principles for classification of organic peroxides

Note: This section refers only to those properties of organic peroxides which are decisive for their classification. A flow chart, presenting the classification principles in the form of a graphically arranged scheme of questions concerning the decisive properties together with the possible answers, is given in figure 2.5.1 in chapter 2.5 of the United Nations *Recommendations on the Transport of Dangerous Goods*. These properties shall be determined experimentally. Suitable test methods with pertinent evaluation criteria are given in the *Manual of Tests and Criteria*, part II.

Part 2 – Classification

- 2.5.3.3.1** Any organic peroxide formulation shall be regarded as possessing explosive properties when, in laboratory testing, the formulation is liable to detonate, to deflagrate rapidly or to show a violent effect when heated under confinement.
- 2.5.3.3.2** The following principles apply to the classification of organic peroxide formulations not listed in 2.5.3.2.4:
- .1 Any organic peroxide formulation which can detonate or deflagrate rapidly, as packaged for transport, is prohibited from transport in that packaging under class 5.2 (defined as ORGANIC PEROXIDE TYPE A);
 - .2 Any organic peroxide formulation possessing explosive properties and which, as packaged for transport, neither detonates nor deflagrates rapidly, but is liable to undergo a thermal explosion in that package, shall bear an “EXPLOSIVE” subsidiary hazard label (Model No. 1, see 5.2.2.2.2). Such an organic peroxide may be packaged in amounts of up to 25 kg unless the maximum quantity has to be limited to a lower amount to preclude detonation or rapid deflagration in the package (defined as ORGANIC PEROXIDE TYPE B);
 - .3 Any organic peroxide formulation possessing explosive properties may be transported without an “EXPLOSIVE” subsidiary hazard label when the substance as packaged (maximum 50 kg) for transport cannot detonate or deflagrate rapidly or undergo a thermal explosion (defined as ORGANIC PEROXIDE TYPE C);
 - .4 Any organic peroxide formulation which, in laboratory testing:
 - .1 detonates partially, does not deflagrate rapidly and shows no violent effect when heated under confinement; or
 - .2 does not detonate at all, deflagrates slowly and shows no violent effect when heated under confinement; or
 - .3 does not detonate or deflagrate at all and shows a medium effect when heated under confinement
 is acceptable for transport in packages of not more than 50 kg net mass (defined as ORGANIC PEROXIDE TYPE D);
 - .5 Any organic peroxide formulation which, in laboratory testing, neither detonates nor deflagrates at all and shows low or no effect when heated under confinement is acceptable for transport in packages of not more than 400 kg/450 L (defined as ORGANIC PEROXIDE TYPE E);
 - .6 Any organic peroxide formulation which, in laboratory testing, neither detonates in the cavitated state nor deflagrates at all and shows only a low or no effect when heated under confinement as well as low or no explosive power may be considered for transport in IBCs or tanks (defined as ORGANIC PEROXIDE TYPE F); for additional provisions see 4.1.7 and 4.2.1.13;
 - .7 Any organic peroxide formulation which, in laboratory testing, neither detonates in the cavitated state nor deflagrates at all and shows no effect when heated under confinement nor any explosive power shall be exempted from class 5.2, provided that the formulation is thermally stable (self-accelerating decomposition temperature is 60°C or higher for a 50 kg package) and for liquid formulations diluent type A is used for desensitization (defined as ORGANIC PEROXIDE TYPE G). If the formulation is not thermally stable or a diluent other than type A is used for desensitization, the formulation shall be defined as ORGANIC PEROXIDE TYPE F.
- 2.5.3.4** Temperature control provisions
- 2.5.3.4.0** The properties of some organic peroxides require that they be transported under temperature control. Control and emergency temperatures for currently assigned organic peroxides are shown in the list 2.5.3.2.4. The controlled temperature provisions are given in chapter 7.3.7.
- 2.5.3.4.1** The following organic peroxides shall be subjected to temperature control during transport:
- .1 organic peroxides type B and C with a SADT $\leq 50^{\circ}\text{C}$;
 - .2 organic peroxides type D showing a medium effect when heated under confinement* with a SADT $\leq 50^{\circ}\text{C}$ or showing a low or no effect when heated under confinement with a SADT $\leq 45^{\circ}\text{C}$; and
 - .3 organic peroxides types E and F with a SADT $\leq 45^{\circ}\text{C}$.
- 2.5.3.4.2** Test methods for determining the SADT are given in the *Manual of Tests and Criteria*, part II, section 28. The test selected shall be conducted in a manner which is representative, both in size and material, of the package to be transported.
- 2.5.3.4.3** Test methods for determining the flammability are given in the *Manual of Tests and Criteria*, part III, subsection 32.4. Because organic peroxides may react vigorously when heated, it is recommended to determine their flashpoint using small sample sizes such as described in ISO 3679.

* As determined by test series E as prescribed in the *Manual of Tests and Criteria*, part II.

2.5.3.5 Desensitization of organic peroxides

- 2.5.3.5.1** In order to ensure safety during transport, organic peroxides are in many cases desensitized by organic liquids or solids, inorganic solids or water. Where a percentage of a substance is stipulated, this refers to the percentage by mass, rounded to the nearest whole number. In general, desensitization shall be such that, in case of spillage or fire, the organic peroxide will not concentrate to a dangerous extent.
- 2.5.3.5.2** Unless otherwise stated for the individual organic peroxide formulation, the following definitions apply for diluents used for desensitization:
- .1** Diluents type A are organic liquids which are compatible with the organic peroxide and which have a boiling point of not less than 150°C. Type A diluents may be used for desensitizing all organic peroxides.
 - .2** Diluents type B are organic liquids which are compatible with the organic peroxide and which have a boiling point of less than 150°C but not less than 60°C and a flashpoint of not less than 5°C. Type B diluents may be used for desensitization of all organic peroxides provided that the boiling point is at least 60°C higher than the SADT in a 50 kg package.
- 2.5.3.5.3** Diluents, other than type A or type B, may be added to organic peroxide formulations as listed in 2.5.3.2.4 provided that they are compatible. However, replacement of all or part of a type A or type B diluent by another diluent with differing properties requires that the organic peroxide formulation be re-assessed in accordance with the normal acceptance procedure for class 5.2.
- 2.5.3.5.4** Water may only be used for the desensitization of organic peroxides which are shown in 2.5.3.2.4 or in the statement of approval according to 2.5.3.2.5 as being with water or as a stable dispersion in water.
- 2.5.3.5.5** Organic and inorganic solids may be used for desensitization of organic peroxides provided that they are compatible.
- 2.5.3.5.6** Compatible liquids and solids are those which have no detrimental influence on the thermal stability and hazard type of the organic peroxide formulation.